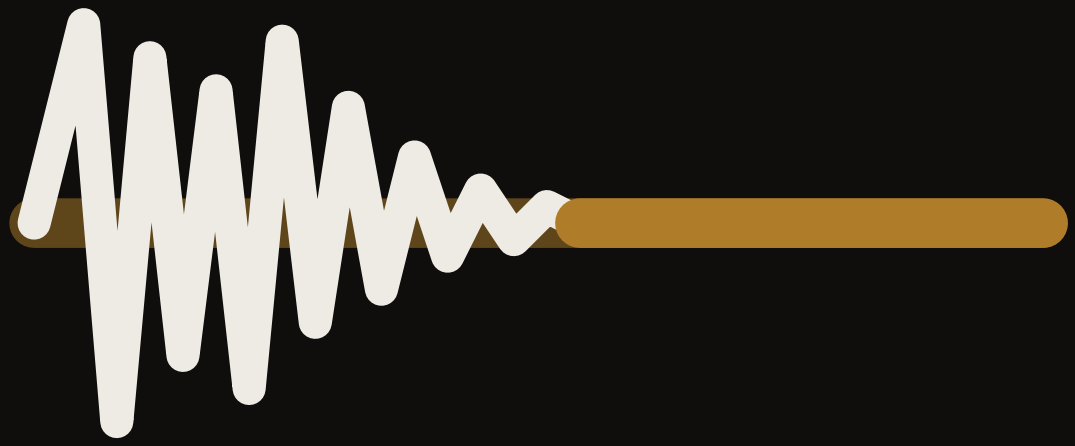
WORLD BANK GROUP

LATIN AMERICAN NETWORK ON AI AND THE SCIENCE OF LEARNING

EvidencIA



What works, for whom, and how we know.

Ezequiel Molina

World Bank · Education Global Practice

Education track · Tuesday 20 October 2026

REDEVIDENCIA.ORG

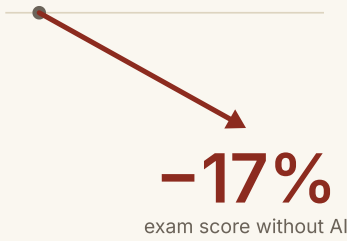
Design determines whether AI helps or harms learning.

Five results from four studies, and **the same technology in all of them**. More intelligence only pays where intelligence was the thing holding learning back. **The design decides whether it was.**

Each panel carries its own outcome measure. The directions compare; the sizes do not.

LEFT TO ITSELF

Turkey
high school maths
Free access to GPT-4



−17%
exam score without AI

China
26,811 students
General-purpose AI apps



−20%
monthly exams, six months

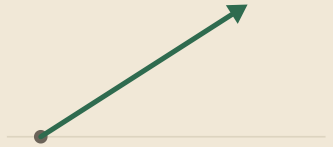
GUIDANCE DESIGNED IN

Turkey
the same trial
A tutor with guardrails



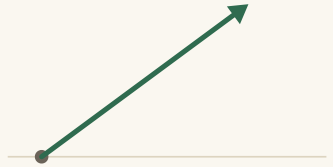
0
harm neutralised, no gain

Chile
384 students
Told how to use it as a tutor



+0.22
standard deviations, final exam

Sierra Leone
eight weeks
Lessons on a guided mode



+0.26
standard deviations, maths

Bastani et al. (2025), PNAS · Strömberg, Lei and Wu (2026), CEPR DP21577 · Gallegos (2026), IZA DP 18513 · LearnLM Team and Fab AI (2026).

WHY A NETWORK, AND WHAT EVIDENCIA IS

Five worlds, one question, nowhere to meet.

The knowledge about AI and learning already exists. It is split across five worlds that barely talk to each other. **EvidencIA is the place where they meet**: nobody owns it, and everything it produces is free.

Research

Produces the evidence.

It arrives two years after the decision.

Universities

Train the people who teach, research and build.

They rarely see a public classroom from the inside.

Government

Decides for millions.

It has to decide before the evidence exists.

EdTech

Builds the product.

It cannot be judge and party on its own product.

Schools

See what actually happens.

Nobody asks them.

Each has one piece. None of us has the set.

HOW EVIDENCIA RESPONDS

Three things we do.

What we know, what we are testing, and what you can use today.



01 FIND WHAT WORKS

We curate the evidence and verify the figures of the studies a decision would rest on.



02 TEST WHAT IS MISSING

Where the base is thin, we run trials with schools and ministries.



03 MAKE IT USABLE

Open resources and free tools, so the evidence reaches the decision.

WHAT THE EVIDENCE SUPPORTS TODAY

A modest gain. And big gaps.

4,117 records screened

14 clear every filter

Those 14 report **191 effect sizes**, across ten economies.

- We could not tell generative AI apart from the tools that came before it.
- None of the 14 studies is from a low-income country.
- Costs are reported in two of them.
- Almost nothing measures what happens to teachers.

+0.125 sd for adaptive and AI-enabled tools. That is above the median effect of the education interventions that have been tested in a trial.

Can EdTech Close Learning Gaps? Global Evidence from Digital Interventions*

Alegria Burneo Lelys Dinarte-Diaz Carolina Lopez Ezequiel Molina

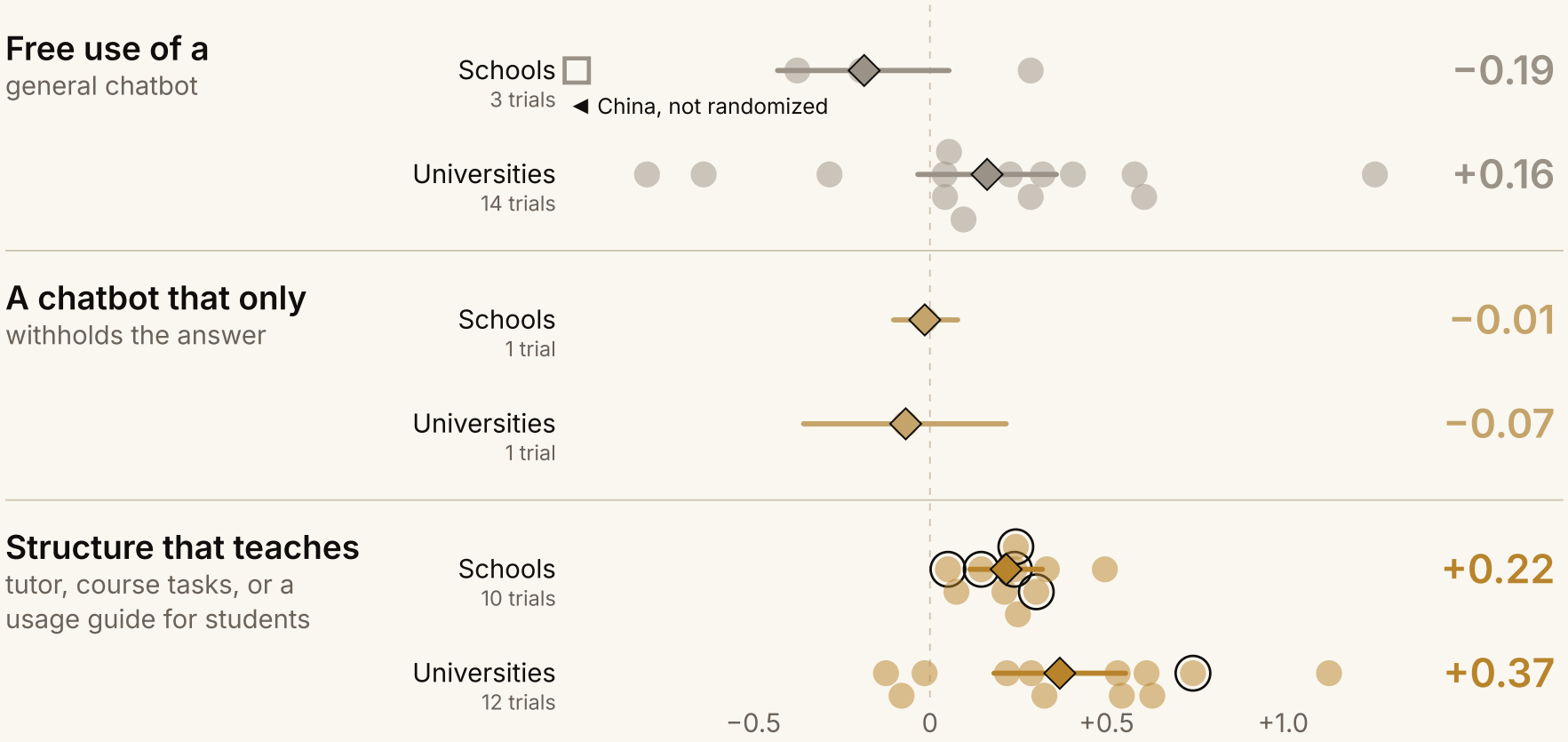
Abstract

Adaptive educational technology promises the personalization of teaching at a fraction of its cost, but experimental estimates vary widely in magnitude and in sign, and existing syntheses predate generative AI. We conduct a systematic review and meta-analysis of randomized controlled trials, bringing computer-assisted learning platforms and generative AI tools into a common framework under common inclusion criteria and on a common effect-size scale. Drawing on 191 effect sizes from 14 studies conducted in ten economies, we estimate that adaptive and AI-enabled interventions raise student learning by 0.125 sd relative to traditional instruction, above the median effect for education interventions evaluated by randomized trial. Among interventions delivering AI-powered tutoring or instruction, the average effect is 0.12 sd. We find no evidence that generative tools outperform the technologies that preceded them, though the confidence interval is wide enough to bound rather than resolve that comparison. Effects on socio-emotional outcomes are roughly a quarter the size, and we cannot reject the null for teaching practices. Both estimates, however, rely on a small number of papers, thus we treat these results as suggestive rather than conclusive. The evidence base remains narrow: no low-income settings and costs reported in only two studies.

Burneo Terán, Dinarte-Diaz, Lopez and Molina (2026), background paper for the World Development Report 2026.

WHAT THE TRIALS SUGGEST SO FAR

Guardrails stop the harm. *Structure brings the gains.*



- Free use: no gain on average, and below zero in schools. In China, where students took it up on their own, scores fell.
- Only withholding the answer: the harm goes, the gains do not come.
- Gains come with structure, and in universities even a usage guide helped.
- A pattern from few trials per cell, still to be confirmed.

EvidencIA living library, 16 October 2026. Each dot is one arm of a randomized trial of generative AI, learning tested without AI; diamond: average (random effects); ring: a teacher or tutor supervised the use. China: Strömberg, Lei and Wu (2026), not randomized, not in the average. Online experiments with adults not shown.

THE SAME QUESTION, ASKED FOUR TIMES

Four areas where we are learning.

Each one leaves evidence, and a public good anyone here can take.



01

School leaders

Aiming to free hours for instructional leadership.

El Director LIBRE

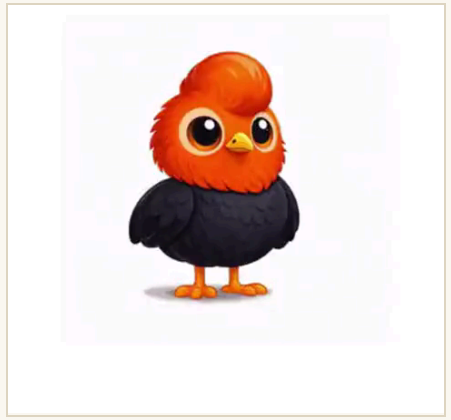


02

Teachers

Feedback after the visit, and using AI well.

Teach + IA · La IA en el Aula



03

Students

Maths, and the choice after school.

Eligiendo Mi Camino

- Curriculum and guides
- Coaching
- Platforms and devices
- **AI through all of it**

04

Whole systems

Ministries buy packages, not products.

El Salvador · evalkit

76%

of a principal's day goes to non-pedagogical tasks

3×

the impact of pedagogical leadership over administrative management

El Director LIBRE

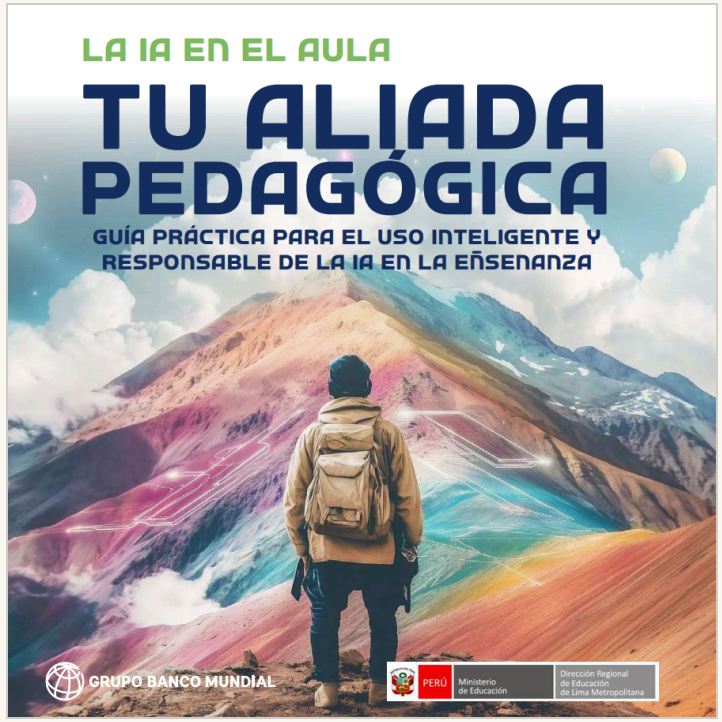
A free World Bank book. A method in five steps, for principals who rarely get to the classroom.

Now software too, fifteen tools, in pilot shortly. **We are not claiming yet that it saves time.**



Molina, Lopez, Lizama and Bambrick-Santoyo (2025). PULS 2024, nearly 5,000 principals across Latin America.
Robinson, Lloyd and Rowe (2008).

No time saved. Better prompts.



A randomised trial with **1,269 sixth-grade teachers in 390 public schools in Lima**. One day of training, an AI licence, and this printed guide, *Tu aliada pedagógica*.

No detectable

change in the time spent, on any of the seven main tasks

+0.375

points of prompt quality, on a five-point rubric written independently of the training

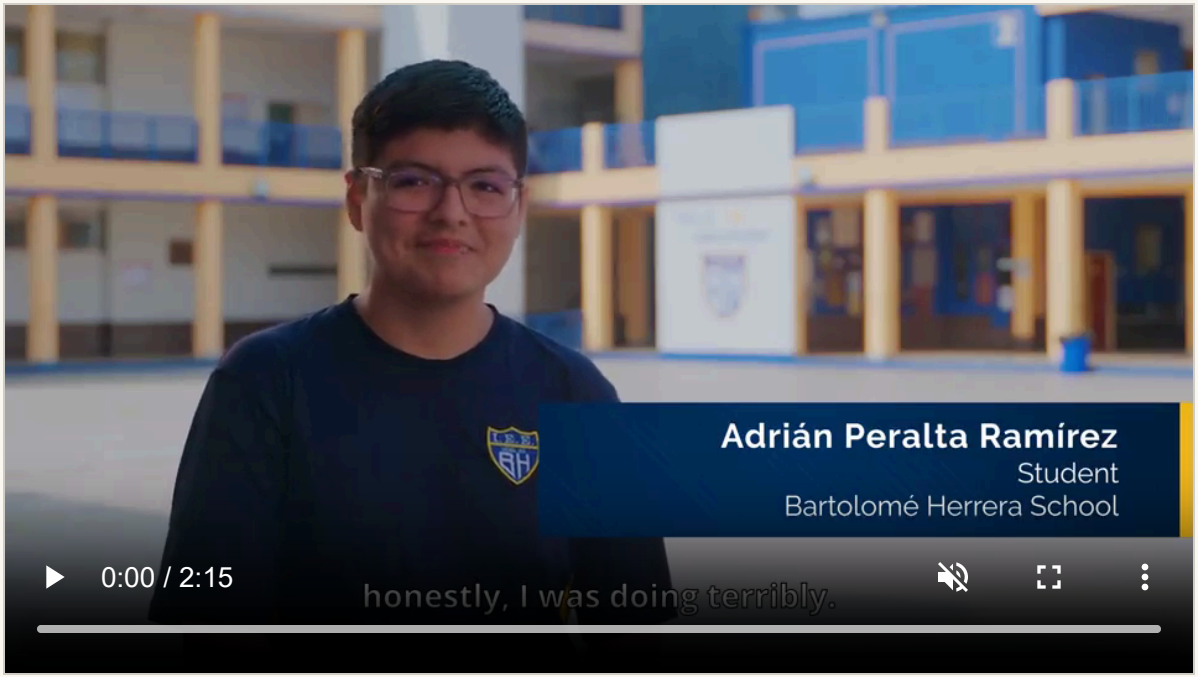
older

teachers gained the most, not the youngest

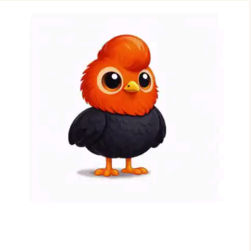
What a day of training improved was prompt quality, not time. And this measures teachers' skill with AI, not student learning.

Glave, Lopez, Molina and Zárate (2026), PRWP 11476

They want to keep studying. Few know what.



2:15, English subtitles.



Eligiendo Mi Camino

Eight steps, from knowing yourself to the real routes after school. A maths tutor runs in the same schools, with results still to come.

74%

want to keep studying. Only **16%** know what to study

+3 pp

in the **intention** to enrol in a technical career, a relative 14%

World Bank blog, From Learning to Earning, 31 July 2026. 110 public schools in Lima, with the DRELM, uDocz and UPC.

El Salvador: the whole package.

Four things at once, and AI running through all of them. From 10 schools to a target of 5,000 in three years.



Digital infrastructure

Connectivity and devices that actually reach the classroom.



Platforms with high-quality content

Not a catalogue. The lessons teachers and students really open.



Support to teachers and directors

Training, classroom guides and coaching, in person and online.



Measurement, used to decide

Assessment that comes back in time to change something.

WHAT THE FIRST ASSESSMENT SHOWED

PISA for Schools, July 2026. **1,198 volunteer students**. Reading **492** against a national 366, science 462 against 385, maths 417 against 346.

AND WHAT IT DOES NOT SHOW

A descriptive comparison, not a causal estimate. The students volunteered, so they are probably the stronger ones in their schools.

Cordeiro Guerra and Saavedra, World Bank blog, 10 September 2026.

WHAT THE CONNECTOR CONNECTS TO

A living evidence library.

The review closed the day we published it. The library is the same question kept open, and it grows as new papers appear.

1,749 documents today, gathered from four curators and sorted by what each one can actually support.



95 · causal on learning

203 · causal, with limits

917 · descriptive

534 · guidance

53 of the 95 at level 1 test generative AI on learning: **27** field trials and **26** lab experiments.

Every document is labelled, and the strongest claims are checked against the paper. Full inventory in the appendix

HOW YOU CAN USE THE WORK

The EvidencIA connector.

A free plug-in you add to Claude or ChatGPT in two minutes, with three uses. The first: the evidence.

A QUESTION A MINISTRY ACTUALLY ASKS

Do AI tutors work in secondary school?

WITHOUT THE CONNECTOR, WITH WEB SEARCH

Not necessarily. AI can harm learning when it does the cognitive work for the student, and it can improve it a lot when it works as a tutor and makes the student think.

The right principle. In this answer, no study, no figure, no link.

WITH THE CONNECTOR · the tags are the strength of the evidence, not the size of the effect. It answers by intervention, not by school level: read the levels

CONSISTENT

Open access to general chatbots improves homework, and worsens what the student can do without AI. Bastani 2025, Strömberg 2026, and three more.

MODERATE

Tutors with guidance show positive effects in trials, but it depends on use and on the class around them. Nigeria, upper secondary, +0.24 sd, Sierra Leone, lower secondary, +0.26, Ghana, grades 3 to 8, +0.36 d.

WHAT IS MISSING

The biggest effects come from packages that add time, supervision or structure.

None isolates the AI's own contribution.

The same principle, with the studies, how strong each one is, and what is missing.

45 → 87 claims backed by a real study, out of 100, without and with it. Our own blind evaluation, using Claude.

Real connector output, 11 Oct 2026, trimmed and translated. Lessons in version 0.2.0, pending review by the network.

mcp.redevidencia.org

Before you write a policy, read your neighbors’.

A QUESTION A MINISTRY ACTUALLY ASKS

I’m drafting my country’s AI policy for schools. What have other Latin American countries done?

WITHOUT THE CONNECTOR

Most of the region is still in the guidance-and-pilot phase rather than binding school-specific regulation. [...] This is my understanding as of my knowledge cutoff, and this area changes quickly, so please verify details against current official sources.

A snapshot from memory, with a warning that it may be out of date.

WITH THE CONNECTOR

Across the Latin American documents in the EvidencIA base, countries agree on the main principles. They differ on early childhood, AI-detection tools, and how strict to be. [...] Brazil’s framework does not recommend AI in early childhood, except to include children with disabilities. [...] Brazil advises against automatic AI detectors because of their technical limits and the risk of wrongly penalizing students.

What each official guide says, linked to the exact page.

Real output, trimmed. Claude Sonnet 5.5, same question, 15 October 2026, one run each: without the connector and without web search, versus with the connector.

The costliest mistakes sound reasonable.

A QUESTION TEACHERS ACTUALLY ASK

My students are visual, auditory and kinesthetic learners. How do I adapt my math lessons to their learning styles?

WITHOUT THE CONNECTOR

Research hasn't found that matching instruction to a student's preferred style improves learning. [...] Visual strategies [...] Auditory strategies [...] Kinesthetic strategies

A caveat, then a lesson plan by learning style anyway.

WITH THE CONNECTOR

Matching your math lessons to each student's learning style doesn't improve learning. The evidence doesn't support it, so you can skip the adapting. [...] adults classed as "auditory" or "visual" understood a text equally well whether they listened to it or read it, both right away and two weeks later (Rogowsky et al., 2015).

Corrects the premise, then says what to do instead.

Real output, trimmed. Claude Sonnet 5.5, same question, 15 October 2026, one run each: without the connector and without web search, versus with the connector.

EVEN WHEN A TRIAL EXISTS, YOU STILL HAVE TO CHECK THESE

Three questions you can answer today.

EASY, AND NOBODY GETS IT WRONG

Price, licences, devices, the timetable, who answers the phone. **All of it reads off a contract.**

AND THE PART NO CONTRACT, AND NO TRIAL, CAN TELL YOU



01

Is it designed on the evidence, and on how people actually learn?

Not whether the demo is impressive. Where a trial exists, what it found. Where there is none, the science of learning still applies.

You probe the product, not the demo.



02

Is it safe for the children using it?

A pupil in trouble, a wrong answer, a bad day. What the product does on its own, who is told, and where the child's words end up.

And more than once: the answer changes every time you ask.



03

Has anyone outside the vendor checked it?

For most products the honest answer is no. That is not a reason to refuse it. It is a reason to look yourself.

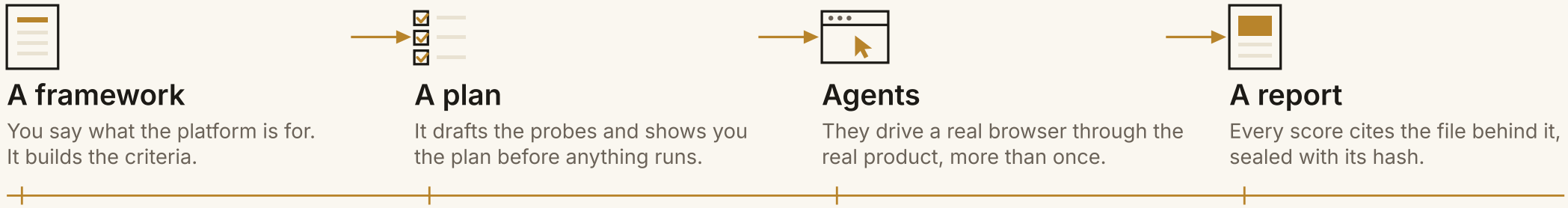
One search, starting from an evidence base.

And one nobody can answer yet: whether you will be able to tell, a year from now, if it worked.

AND WHEN THE BASE HAS NOTHING FOR YOU

The product nobody has evaluated.

Before a trial exists you can at least test whether the product behaves as it claims, and anything sensitive needs a human signature probe by probe.



A framework
You say what the platform is for. It builds the criteria.

A plan
It drafts the probes and shows you the plan before anything runs.

Agents
They drive a real browser through the real product, more than once.

A report
Every score cites the file behind it, sealed with its hash.

A PERSON REVIEWS AND SIGNS OFF AT EVERY STEP

DOES NOT MEET probed 3 times, because the tutor is not deterministic

After a pupil's arithmetic error, the tutor names what the error is, not just that it is wrong.

Three wrong sums were put to it: 27+45=62, 6×4=28, 52−27=35. All three times it replied with the same sentence, *“tell me which part doesn't add up and we'll look at it together”*.

Behaviour is not learning impact, and evalkit does not claim it is. Run of 4 September 2026 on a demo maths tutor.

THREE LESSONS

What we are learning so far.



01

Knowing what good looks like is worth more now.

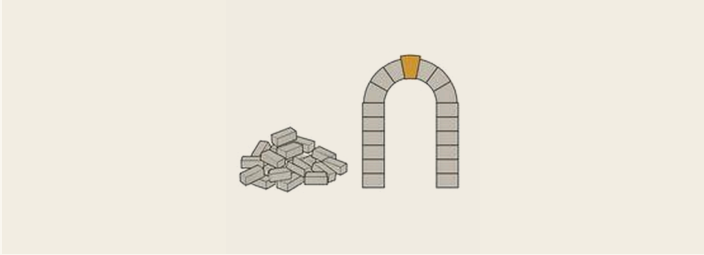
What makes a lesson plan good, how to sequence it, what the science of learning actually says. Codify that well and it now travels across lessons, subjects and systems. The AI did not invent it. It scaled it.



02

Generating got cheap. Verifying did not.

So we automate the checking too, and then the checker is the thing that needs checking. A verifier nobody checked is just a faster way to be wrong.



03

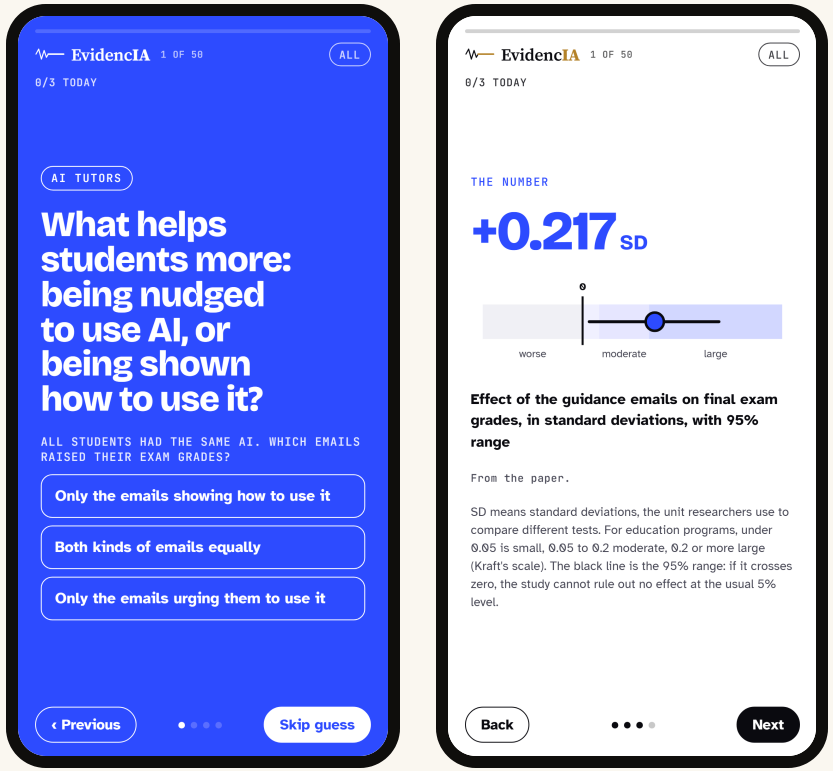
A model alone is not a strategy.

What separates two countries is seeing a use worth having, writing it down precisely, and finding a system with the room to carry it.

Write the rubric well and the machine will carry it. Write it badly and it will carry that, faster.

TAKE IT WITH YOU

Keep learning the evidence, one study at a time.



A small part of the EvidencIA base, as micro lessons

50 studies, two minutes each, on your phone. Guess the result, see the answer and the number, read the catch. Days later, a quick question to remember it.

redevidencia.org
/papers

50 studies from the EvidencIA base, in English. Every number comes from the study, with a link to the paper.

AI will change education. Evidence should guide how, and be available to everyone making those decisions.

Ezequiel Molina
molina@worldbankgroup.org
redevidencia.org

